Edbert Aksama

University of Washington | GEOG 426

Final Project | 8/20/2021

Spatial and Statistical Analysis of Political Affiliation in Washington State.

Abstract

In this report, inferential and descriptive spatial analysis is applied in order to examine the spatial trends of major political affiliation. I specifically look at the spatial distribution of proportion of democratic or republican votes out of total votes for 2020 presidential election in counties in Washington State and examine how the major political affiliation is distributed. This spatial distribution is analyzed in an attempt to describe the political climate of Washington State. Furthermore, a temporal and spatial analysis will also be applied in order to examine temporal trends in an attempt to examine how the political climate has changed over time.

Introduction

Politics have played a large part in the lives of people in America and it has been since the start of the country itself as it has a huge influence on the lives of its citizens. To cast your vote is a critical decision to make as the actions of the people that the citizens voted to power will better or worsen their lives. However, there is a huge divisiveness in the opinions of the people making it a very controversial topic. The nation is divided between Republicans and Democrats.

Therefore, for my final project, I propose to conduct an exploratory analysis of Washington's vote count data that is obtained. My primary objective is to better understand the distribution and characteristics of Washington's major political climate, exploring the spatial locations of intense political affiliation as describing the political climate of an area of a country can hint towards the greater political climate of the country itself. In this analysis, I explore spatial distribution and characteristics of both major political affiliations, identify the presence of clusters of both major political affiliations, identify locations of these clusters, and explore spatial temporal trends of both major political affiliations.

Data

To begin, data obtained from MIT Election Data found at : https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VOQCHQ consists of data regarding vote counts for each political parties for all 3,006 county in the United States from 2000 to 2020. Total amount of votes counted in each county for the given year is also recorded within the data. The data is then filtered down since the area of interest of this analysis is Washington State and is further filtered down to only contain data for the year of 2020 which is the chosen year of analysis. 2020 is chosen year of analysis as it is the most recent year available within the data thus making it more representative of current political climate. Filtered data is then separated according to the two major political affiliations in order to analyze both independently of each other.

Furthermore, data obtained from MIT Election Data goes through a different filter process whereby the data is filtered down since area of interest is Washington thus only displaying vote count data of Washington from 2000 to 2020 which is then separated according to the two major political affiliations in order to apply independent spatial and temporal analysis on both political affiliations.

Washington county shapefile is obtained through Washington's Open Data initiative found at: https://geo.wa.gov/datasets/wadnr::wa-county-boundaries-1/explore?location=47.244239%2C-120.817600%2C7.68 to be joined with filtered and separated datasets.

Methodology and Results

As an exploratory step to provide a context as to the general distribution of political affiliation throughout Washington, I made a choropleth map of both Democrat and Republican layer to visually display the percentage of Democratic or Republican votes in each county which can be seen in Figure 1 and 2.

Proportion of Democratic Votes out of Total Votes for Presidential Candidate in Counties of Washington in 2020

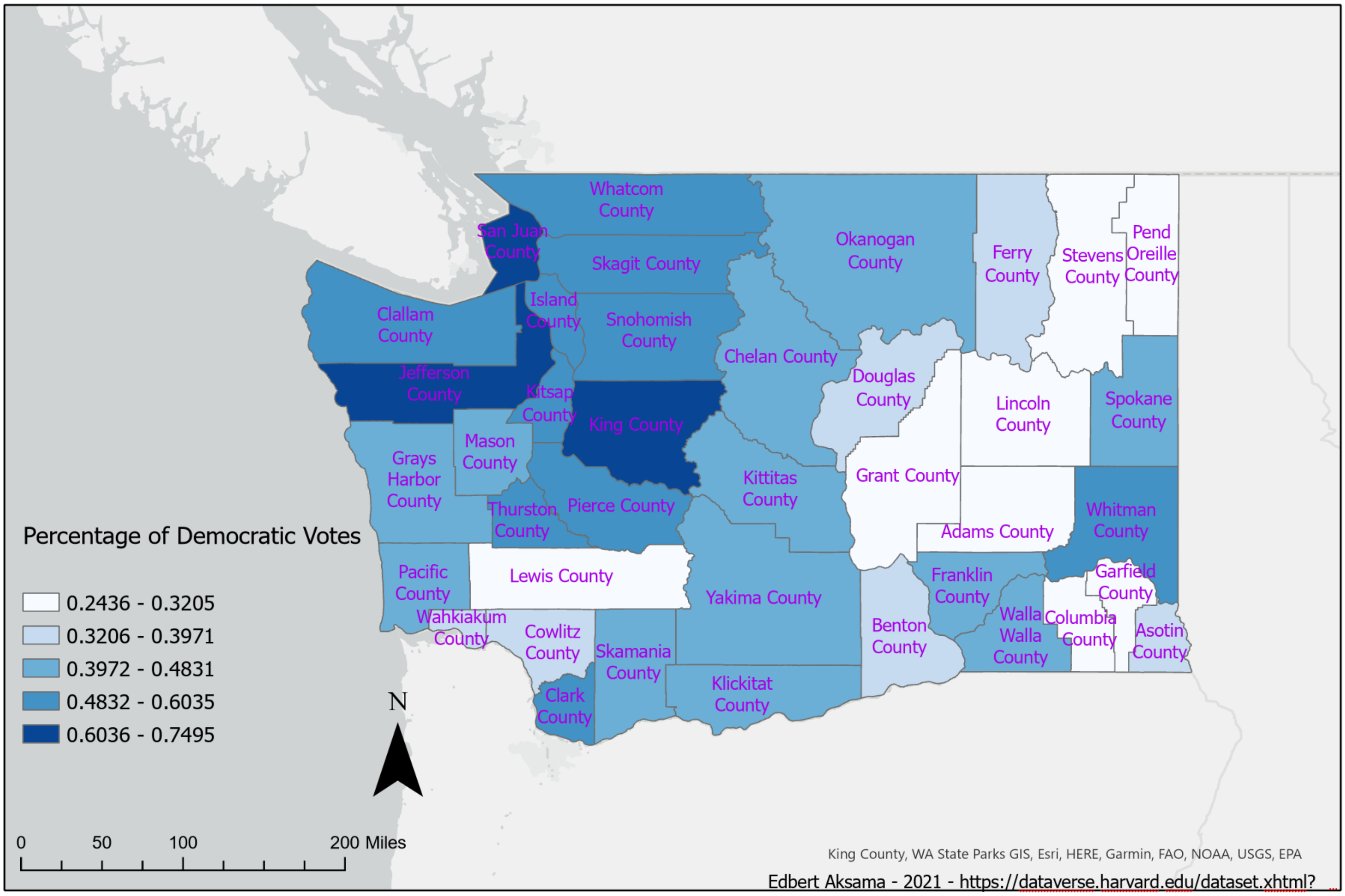

Figure 1 Map displaying percentage of democratic votes out of total votes from each county

Proportion of Republican Votes out of Total Votes for Presidential Candidate in Counties of Washington in 2020

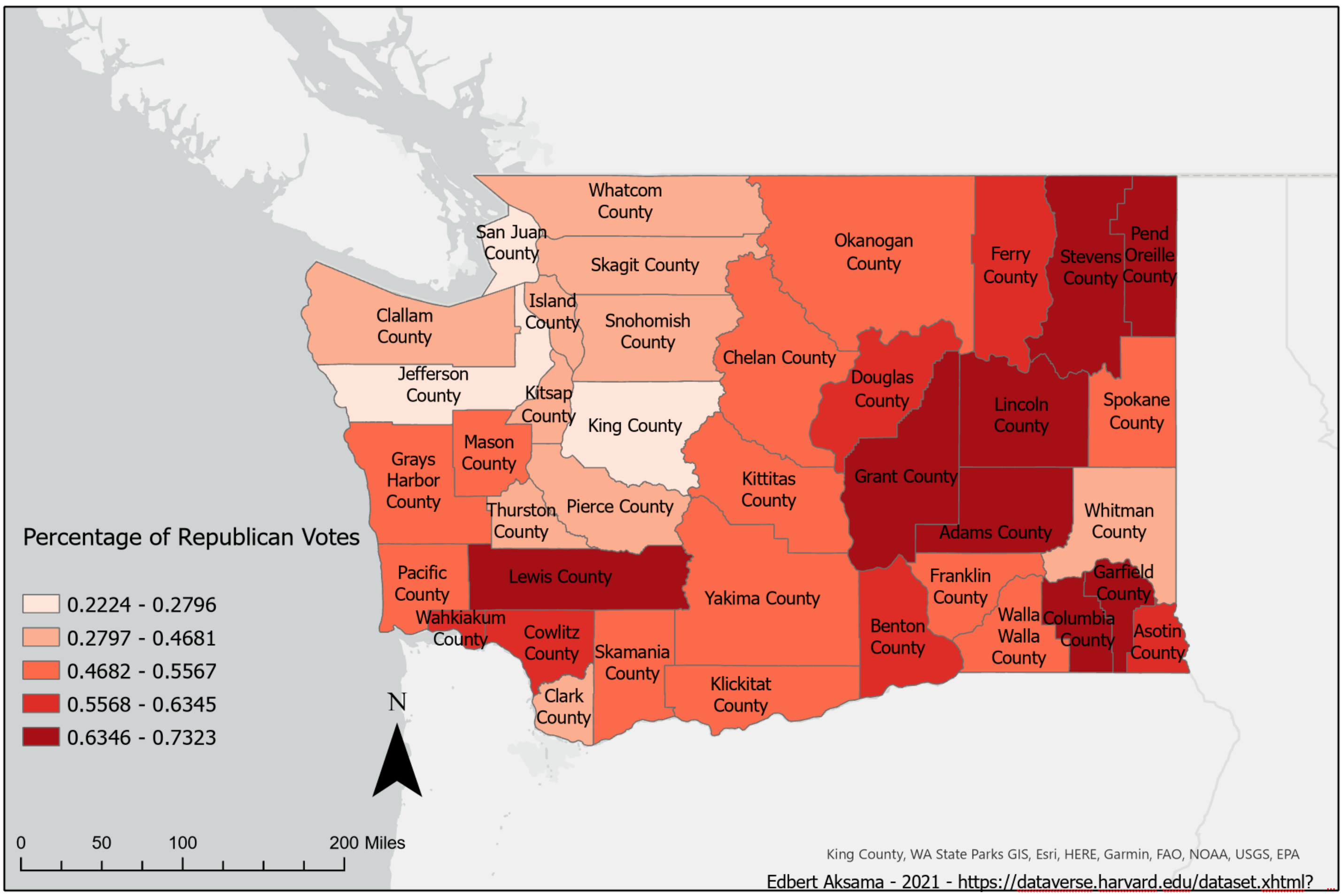

Figure 2 Map displaying percentage of republican votes out of total votes from each county

Both Figure 1 and 2 displays higher proportion of votes for that particular party in darker colors while lower proportion of votes for that particular party in lighter colors. Therefore, it can be seen from the figures that there are more Democratic votes in the western part of Washington State and there are more Republican votes in the eastern part of Washington. The figure displays a clear spatial division between both political parties thus signifying a clustering pattern within the data itself.

Furthermore, I conducted various simple descriptive spatial analysis on both Democrat and Republican layer to examine characteristics of distribution of political affiliation and identify any patterns that might indicate the clustering of political affiliation. In order to apply this analysis, the measuring geographic distribution analysis tool is used to record measures of central tendency such central feature, mean center, median center, and measures of dispersion such as standard distance and standard deviational ellipses, both unweighted and weighted based on candidate votes. Result of the analysis can be seen in Table 1 and weighted statistics for both major political affiliation is displayed in Figure 3 and 4.

	WA		Democratic vote weighted		Republican vote weighted	
mean center	-13439486.1242748°N	5982053.5029224°W	-13537432.832397°N	6007879.47546215°W	-13403935.2685229°N	5972870.27323737°W
median center	-13454383.6934519°N	5978447.40234362°W	-13562716.172496°N	6022720.3196256°W	-13399604.5737765°N	5969458.59751887°W
standard distance center	-13439486.1242748°N	5982053.5029224°W	-13537432.832397°N	6007879.47546215°W	-13403935.2685229°N	5972870.27323737°W
directional distribution center	-13439486.1242748°N	5982053.5029224°W	-13537432.832397°N	6007879.47546215°W	-13403935.2685229°N	5972870.27323737°W
standard distance length(m)	1775863.403		1172676.068		1785759.839	
standard distance area(m^2)	2.50956E+11		1.0943E+11		2.53761E+11	
Directional Distribution Length	1749120.424		1164757.264		1757523.192	

Directional Distribution Area(m^2)	2.2058E+11	1.03507E+11	2.215E+11

Table 1 Coordinate location of mean and median center of WA, Standard Distance, and Directional Distance. Weighted data accounts for proportion of party vote

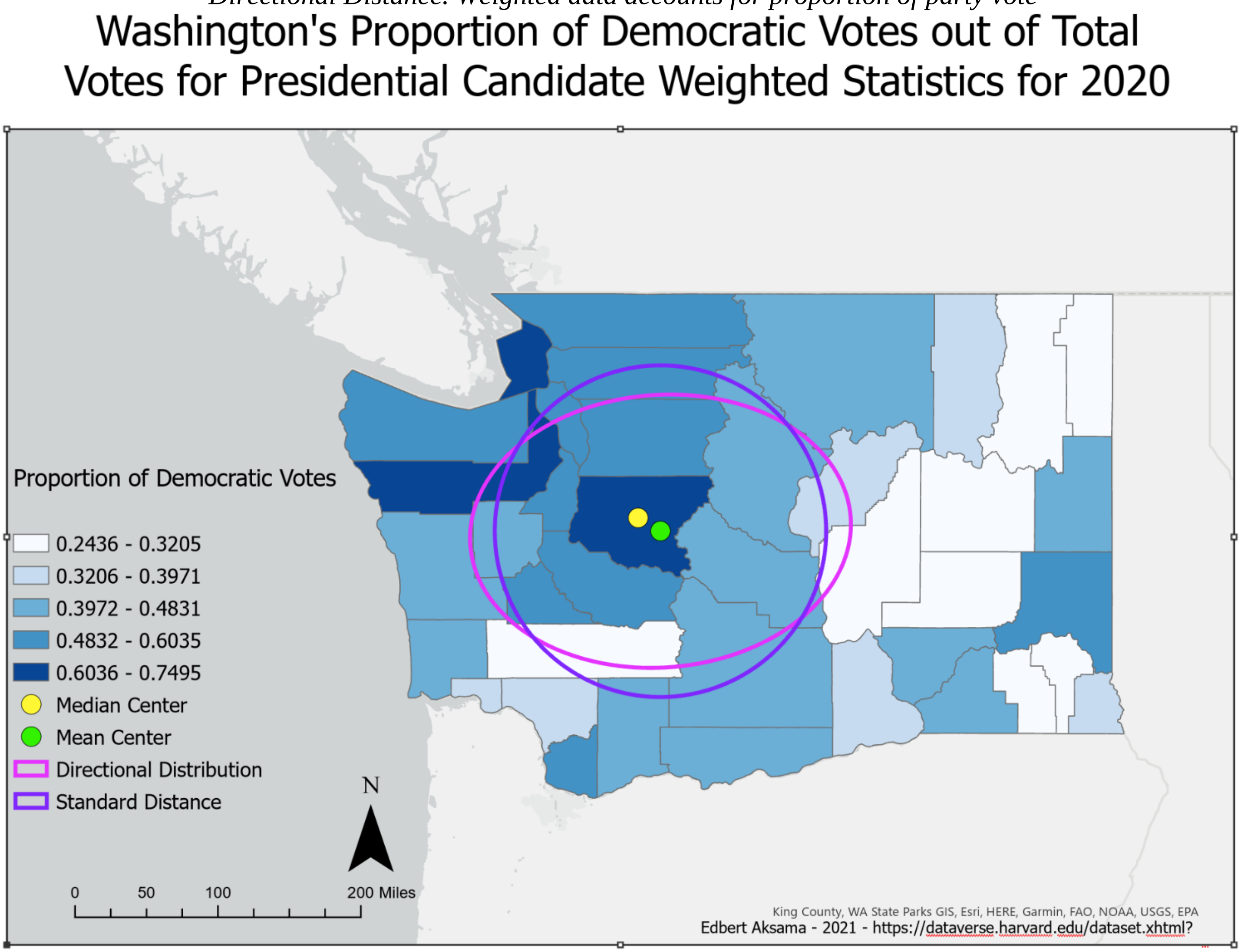

Figure 3 Map from Fig. 1 that highlights weighted center points, Standard Distance, and Directional Distribution.

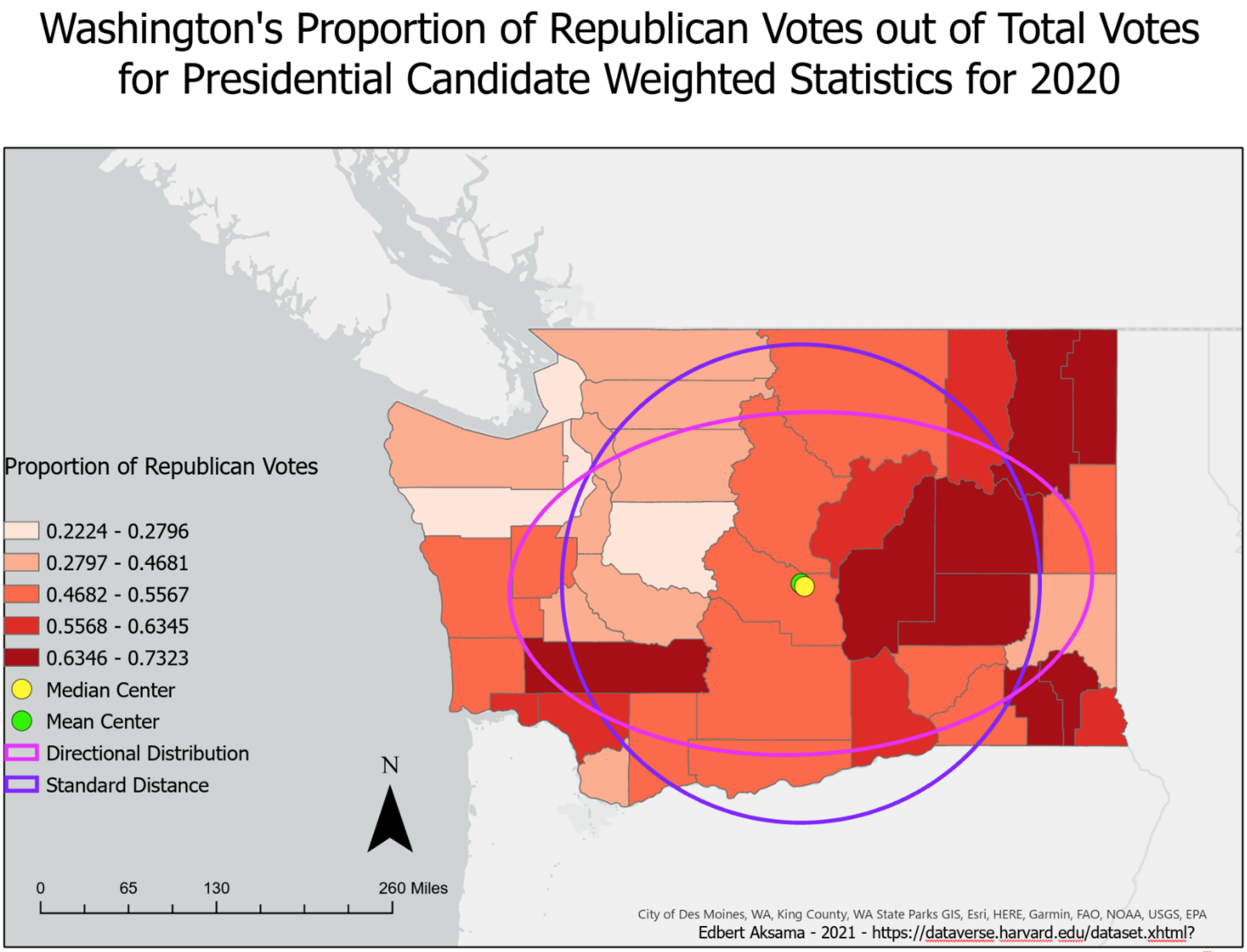

Figure 4 Map from Fig. 2 that highlights weighted center points, Standard Distance, and Directional Distribution.

Table 1 provides specific coordinates of the mean and median centers for the state. These center points only accurately reflect the center location of the state itself since it does not account for proportion of votes for each county. Weighted data that takes this into account is also provided by Table 1. Clustering of high proportion of Democratic votes have shifted the mean and median centers to NW direction while clustering of high proportion of Republican votes have shifted the mean and median center to SE direction. It can also be seen in Figure 3 and 4 that Democrats have a smaller standard distance and standard deviational ellipses than Republicans thus indicating that Republicans are much more dispersed throughout the state than Democrats while Democrats are more concentrated.

It is indicated from the results of previous analyses applied that there is evidence of clustering of political affiliations in Washington state. In order to examine the characteristics of the cluster itself, both the Global Moran's I test and Getis Ord General G analysis is applied. The Global Moran's I test examines whether spatial processes promoting the observed pattern of values is random chance therefore providing a measure of the overall spatial correlation of the dataset by comparing how one object of interest is like others surrounding it. On the other hand, the Getis Ord General G analysis examines whether there is a clustering of values therefore providing a measure of degree of clustering within dataset by comparing each feature to other features. The result of both analyses being applied to both layers can be seen through Table 2.

Democratic Votes	**Moran's I Index**	**Z score**	**P-value**	**Dispersed, Random**
				or Clustered
contiguity edges only	0.1590	3.1646	0.0016	Clustered
contiguity edges and corners	0.1376	2.8186	0.0048	Clustered
Inverse distance	0.2577	4.0340	0.0001	Clustered
Inverse distance squared	0.2669	3.9718	0.0001	Clustered
K_nearest_neighbor (K=10)	0.0553	2.4742	0.0134	Clustered
Republican Votes	**Moran's I Index**	**Z score**	**P-value**	**Dispersed, Random**
				or Clustered
contiguity edges only	0.2021	2.4155	0.0157	Clustered

contiguity edges and corners	0.1775	2.1705	0.0300	Clustered
Inverse distance	0.3117	2.9628	0.0030	Clustered
Inverse distance squared	0.3292	2.9715	0.0030	Clustered
K_nearest_neighbor (K=10)	0.0635	1.7143	0.0865	Clustered
Democratic Votes	**Observed General G**	**Z score**	**P-value**	**Clustering of**
				High or Low
contiguity edges only	0.0595	2.8560	0.0043	High value clustering
contiguity edges and corners	0.0564	2.6062	0.0092	High value clustering
Inverse distance	0.0763	3.6229	0.0003	High value clustering
Inverse distance squared	0.0779	3.5674	0.0004	High value clustering
K_nearest_neighbor (K=10)	0.0511	3.6049	0.0003	High value clustering
Republican Votes	**Observed General G**	**Z score**	**P-value**	**Clustering of**
				High or Low
contiguity edges only	0.0389	2.0554	0.0398	High value clustering
contiguity edges and corners	0.0379	1.9160	0.0554	High value clustering
Inverse distance	0.0444	2.5307	0.0114	High value clustering

Inverse distance squared	0.0452	2.5130	0.0120	High value clustering
K_nearest_neighbor (K=10)	0.0373	2.8990	0.0037	High value clustering

Table 2 Result from Global Moran's I and Getis Ord General G

It can be seen from Table 2 that both statistical test indicates that pattern exhibited by both major political affiliations is clustered in a certain way and is not due to random chance that it exhibits this pattern with only slight differences in results. However, the slight differences in Moran's I indicate that clustering of Republican affiliation is more surrounded by other Republican affiliation than Democratic affiliation and the slight differences in Getis Ord General G indicate that the cluster of Democratic affiliation is more concentrated than clusters of Republican affiliation.

As the characteristics of the clusters are identified, I wanted to identify the specific locations of these clusters and any outliers that the cluster may have. To do so, a local Moran's I is applied to both layers which will compare proportion within counties to others within a certain distance, resulting in the identification of neighborhood clusters and outliers of each crime type which can be seen in Figure 5 and 6.

Clusters and Outliers of Proportion of Democratic Votes out of Total Votes for Presidential Candidate in Washington in 2020

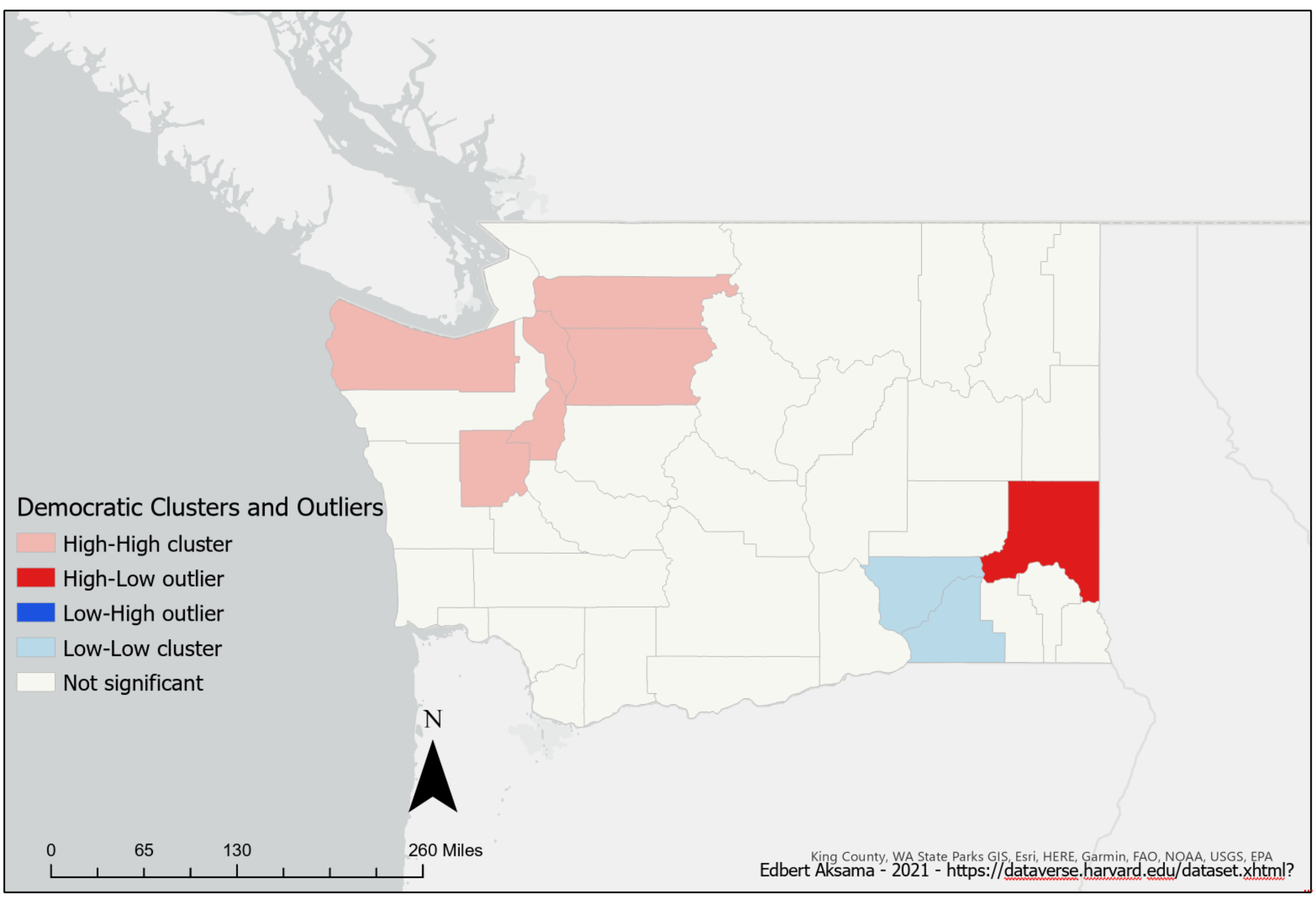

Figure 5 Map highlighting county clusters and outliers of democratic votes

Clusters and Outliers of Proportion of Republican Votes out of Total Votes for Presidential Candidate in Washington in 2020

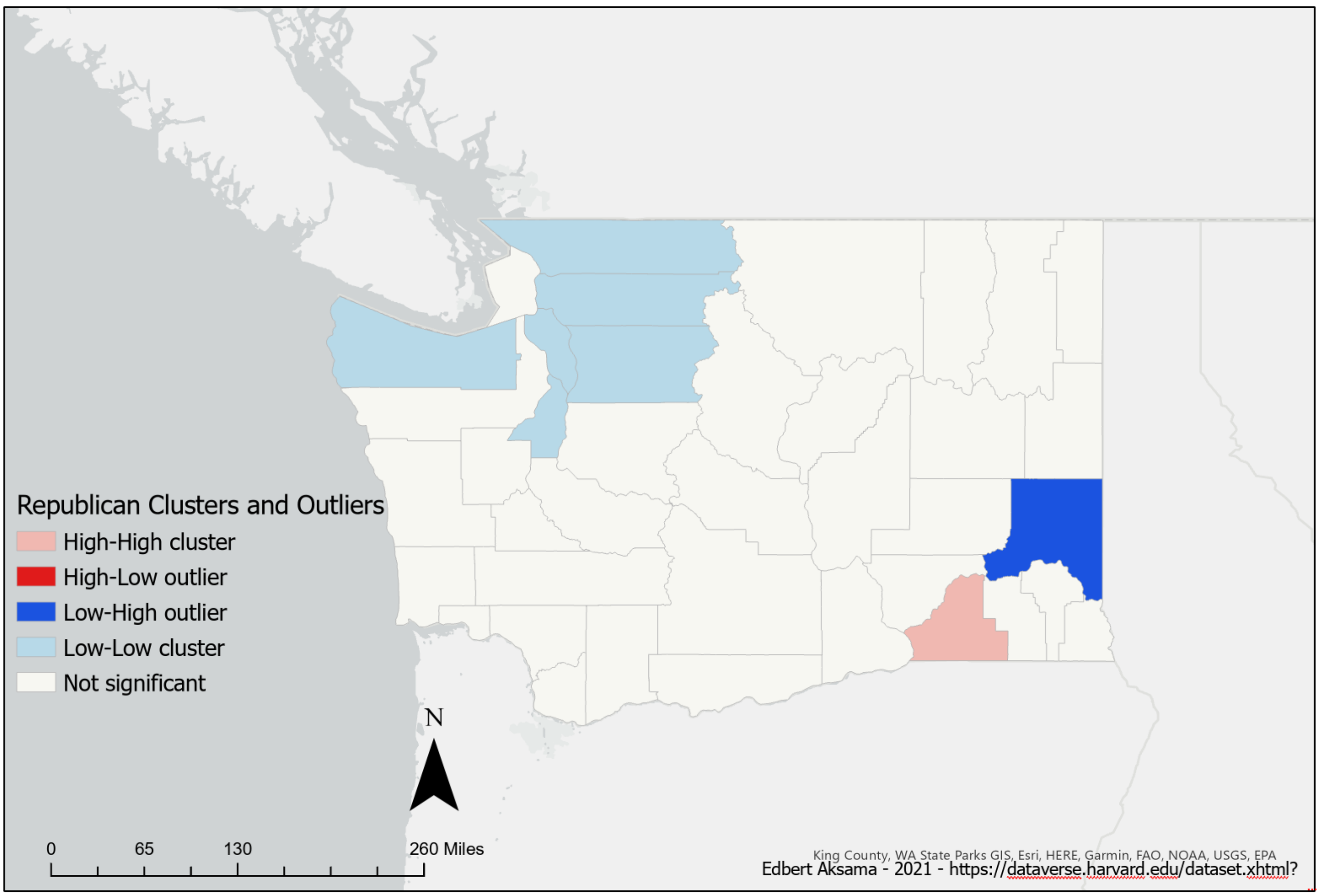

Figure 5 Map highlighting county clusters and outliers of republican votes

Figure 5 highlights statistically significant county clusters and outliers of proportion of Democratic votes within Washington state. It can be seen that there is a high clustering of high Democratic votes within the Northeastern area of the state with no low Democratic vote outliers surrounding the area. Clusters of low Democratic votes can be seen in the Southwestern area of the state with one high Democratic vote outlier in the area. On the other hand, Figure 6 highlights statistically significant county clusters and outliers of proportion of Republican votes within Washington state. It can be seen that there is a clustering of high Republican votes within the Southwestern area of the state with one low republican vote outlier in the area. Clusters of low Republican votes can be seen in the Northeaster area of the state with no high Republican vote outlier within the area. Therefore, both figures are essentially the inverse of each other signifying that concentrations of both major political affiliation does not coexist.

In order to examine how these trends have changed over time, a temporal spatial analysis is applied which requires the creation of a time cube. However, this analysis can only be attempted and did not yield any results, since the time cube itself cannot be created due to the lack of slices of data. Since data for presidential election is only from 2000 to 2020 and the fact that the presidential election only happens every 4 years, only 5 slices of data can be generated by the software.

Conclusion

The result of the analysis indicates that the political climate within Washington state is intensely divided. Through the exploratory analysis, it is discovered more Democrats can be found in the Northeastern area of the state while more Republicans can be found in the South and West side of the state thus signifying clustering of political affiliations. It is also discovered that Republicans are much more dispersed over a large area of the state while Democrats occupy a smaller area of the state and is more concentrated. Democratic clusters are more concentrated over a small area while Republican clusters are found to usually have other Republican oriented counties surrounding them. Although this analysis does not explain as to why this phenomenon occurs, results of this analysis can be combined with other analysis in order explore the driving forces of this phenomenon.